Proof

Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k},\ \vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k},\ \vec{c}=c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$. Then, we have

$$\vec{a},\vec{b},\vec{c} \text{ are coplanar } \Leftrightarrow \left[\vec{a},\vec{b},\vec{c}\right]=0 \Leftrightarrow \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}=0$$

$\Leftrightarrow$ there exist scalars $r,s,t\in\mathbb{R}$,

atleast one of them non-zero such that

$$a_1r+a_2s+a_3t=0,\ b_1r+b_2s+b_3t=0,\ c_1r+c_2s+c_3t=0$$

$\Leftrightarrow$ there exist scalars $r,s,t\in\mathbb{R}$,

atleast one of them non-zero such that $r\vec{a}+s\vec{b}+t\vec{c}=\vec{0}$.

Theorem 6.6

If $\vec{a},\vec{b},\vec{c}$ and $\vec{p},\vec{q},\vec{r}$ are any two systems of three vectors, and if $\vec{p}=x_1\vec{a}+y_1\vec{b}+z_1\vec{c}$, $\vec{q}=x_2\vec{a}+y_2\vec{b}+z_2\vec{c}$, and, $\vec{r}=x_3\vec{a}+y_3\vec{b}+z_3\vec{c}$, then

$$\left[\vec{p},\vec{q},\vec{r}\right]=\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}\left[\vec{a},\vec{b},\vec{c}\right].$$

Note

By theorem 6.6, if $\vec{a},\vec{b},\vec{c}$ are non-coplanar and

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}\neq 0,$$

then the three vectors $\vec{p}=x_1\vec{a}+y_1\vec{b}+z_1\vec{c},\quad \vec{q}=x_2\vec{a}+y_2\vec{b}+z_2\vec{c}$, and, $\vec{r}=x_3\vec{a}+y_3\vec{b}+z_3\vec{c}$ are also non-coplanar.

Example 6.12

If $\vec{a}=-3\hat{i}-\hat{j}+5\hat{k},\ \vec{b}=\hat{i}-2\hat{j}+\hat{k},\ \vec{c}=4\hat{j}-5\hat{k}$, find $\vec{a}\cdot(\vec{b}\times\vec{c})$.

Solution: By the defination of scalar triple product of three vectors,

$$\vec{a}\cdot(\vec{b}\times\vec{c})=\begin{vmatrix} -3 & -1 & 5 \\ 1 & -2 & 1 \\ 0 & 4 & -5 \end{vmatrix}=-3.$$

12th_Maths_EM_Vol1_CH 6 Vector Algebra.indd 235 16-12-2022 14:24:57